Week 8: Database Relationship Analysis using Joins

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Scenario:

The e-commerce company maintains customer, product, order, and payment information in its database. In this week, SQL joins are implemented to combine related data from multiple tables using INNER JOIN, LEFT JOIN, and RIGHT JOIN. Queries are used to identify customers without purchases, products without sales, and orders with their payment details. Customer purchase history and multi-table business reports are also generated to analyse customer spending, order counts, sales, and payment information.

1. INNER JOIN

INNER JOIN returns only matching records from both tables.

1. Display customer details along with their orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM Customer C
  4  INNER JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C001  Arun Kumar                            1001 20-SEP-26         1299
Pending

C002  Aakash                                1002 21-SEP-26          899
Shipped

C003  Rahul Raj                             1003 22-SEP-26         1599
Delivered


CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C001  Arun Kumar                            1004 23-SEP-26          699
Pending

C004  Divya Kumar                           1005 24-SEP-26          450
Shipped

C005  Karthik S                             1006 24-SEP-26          750
Pending
```

```
CUSTO CUSTOMER_NAME                    ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C002  Aakash                               1007 25-SEP-26         1200
Delivered

C003  Rahul Raj                            1008 25-SEP-26          550
Shipped

C005  Karthik S                            1009 26-SEP-26          999
Pending


CUSTO CUSTOMER_NAME                    ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C004  Divya Kumar                          1010 24-SEP-26          550
Shipped

C004  Divya Kumar                          1011 24-SEP-26          500
Delivered

C005  Karthik S                            1012 24-SEP-26          550
Pending


CUSTO CUSTOMER_NAME                    ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C002  Aakash                               1013 28-SEP-26          499
Delivered

C003  Rahul Raj                            1014 29-SEP-26          199
Delivered

C004  Divya Kumar                          1015 30-SEP-26          229
Delivered
```

```
CUSTO CUSTOMER_NAME                    ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
ORDER_STATUS
--------------------
C005  Karthik S                            1016 01-OCT-26          249
Delivered

C001  Arun Kumar                           1017 27-SEP-26          449
Delivered


17 rows selected.
```

2. Display order details with payment information

```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT,
  2         P.PAYMENT_MODE, P.PAYMENT_STATUS, P.TRANSACTION_AMOUNT
  3  FROM Orders O
  4  INNER JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1001 20-SEP-26         1299 UPI                  Completed
              1299

      1002 21-SEP-26          899 Credit Card          Completed
               899

      1003 22-SEP-26         1599 Debit Card           Completed
              1599


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1004 23-SEP-26          699 UPI                  Pending
               699

      1005 24-SEP-26          450 Credit Card          Completed
               450

      1006 24-SEP-26          750 Cash                 Pending
               750
```

```
  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1007 25-SEP-26         1200 UPI                  Completed
              1200

      1008 25-SEP-26          550 Debit Card           Completed
               550

      1009 26-SEP-26          999 UPI                  Pending
               999


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1010 24-SEP-26          550 Credit Card          Failed
               550

      1011 24-SEP-26          500 UPI                  Completed
               450

      1012 24-SEP-26          550 Credit Card          Completed
               550


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1013 28-SEP-26          499 UPI                  Completed
               499

      1014 29-SEP-26          199 Debit Card           Completed
               199

      1015 30-SEP-26          229 Credit Card          Completed
```

```
      1015 30-SEP-26          229 Credit Card          Completed
               229


  ORDER_ID ORDER_DAT TOTAL_AMOUNT PAYMENT_MODE         PAYMENT_STATUS
---------- --------- ------------ -------------------- --------------------
TRANSACTION_AMOUNT
------------------
      1016 01-OCT-26          249 UPI                  Completed
               249

      1017 27-SEP-26          449 Cash                 Pending
               449


17 rows selected.
```

3. Retrieve products purchased by customers

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID, P.PRODUCT_NAME,
  2         OD.QUANTITY, OD.PRICE
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  ORDER BY C.CUSTOMER_NAME, O.ORDER_ID;

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Aakash                              1002
Crystal Clear Case
         1        899

Aakash                              1007
Crystal Clear Case
         1       1200

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Aakash                              1013
Heavy Duty Case
         1        499

Arun Kumar                          1001
Soft Silicone Case
```

```
Arun Kumar                          1001
Soft Silicone Case

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1        301

Arun Kumar                          1001
Clear Shock Case
         2        499

Arun Kumar                          1004

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Luxury Designer Case
         1        699

Arun Kumar                          1017
Personal Photo Case
         1        449


CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Divya Kumar                         1005
Clear Shock Case
         1        450
```

```
CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Divya Kumar                         1011
Luxury Gold Case
         1        500

Divya Kumar                         1015
Classic Black Case

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1        229

Karthik S                           1006
Soft Silicone Case
         2        375

Karthik S                           1009

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Luxury Designer Case
         1        999

Karthik S                           1012
Collage Photo Case
         1        550
```

```
CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
Karthik S                           1016
Slim Basic Case
         1        249

Rahul Raj                           1003
Floral Printed Case
         2      799.5

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------

Rahul Raj                           1008
Floral Printed Case
         2       1000

Rahul Raj                           1014
Matte Basic Case

CUSTOMER_NAME                  ORDER_ID
------------------------------ ----------
PRODUCT_NAME
--------------------------------------------------------------
  QUANTITY      PRICE
---------- ----------
         1        199


18 rows selected.
```

2. LEFT JOIN

4. Display all customers, including those without orders

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY C.CUSTOMER_ID;

CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
C001  Arun Kumar                            1001 20-SEP-26         1299
C001  Arun Kumar                            1004 23-SEP-26          699
C001  Arun Kumar                            1017 27-SEP-26          449
C002  Aakash                                1002 21-SEP-26          899
C002  Aakash                                1007 25-SEP-26         1200
C002  Aakash                                1013 28-SEP-26          499
C003  Rahul Raj                             1003 22-SEP-26         1599
C003  Rahul Raj                             1008 25-SEP-26          550
C003  Rahul Raj                             1014 29-SEP-26          199
C004  Divya Kumar                           1005 24-SEP-26          450
C004  Divya Kumar                           1011 24-SEP-26          500

CUSTO CUSTOMER_NAME                   ORDER_ID ORDER_DAT TOTAL_AMOUNT
----- ------------------------------ ---------- --------- ------------
C004  Divya Kumar                           1010 24-SEP-26          550
C004  Divya Kumar                           1015 30-SEP-26          229
C005  Karthik S                             1006 24-SEP-26          750
C005  Karthik S                             1009 26-SEP-26          999
C005  Karthik S                             1012 24-SEP-26          550
C005  Karthik S                             1016 01-OCT-26          249
C006  Priya Sharma

18 rows selected.
```

5. Find customers without any purchases

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME
  2  FROM Customer C
  3  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  WHERE O.ORDER_ID IS NULL;

CUSTO CUSTOMER_NAME
----- ------------------------------
C006  Priya Sharma
```

6. Display all products, including products with no sales

```
PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
Military Grade Case


P000029
Everyday Basic Case



PRODUCT
-------
PRODUCT_NAME
------------------------------------------------------------
  ORDER_ID   QUANTITY
---------- ----------
P000030
Simple White Case



36 rows selected.
```

3. RIGHT JOIN

7. Display all orders with customer information

```
SQL> SELECT O.ORDER_ID, O.ORDER_DATE, O.TOTAL_AMOUNT,
  2         C.CUSTOMER_ID, C.CUSTOMER_NAME
  3  FROM Customer C
  4  RIGHT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTO CUSTOMER_NAME
---------- --------- ------------ ----- -------------------------
      1001 20-SEP-26         1299 C001  Arun Kumar
      1002 21-SEP-26          899 C002  Aakash
      1003 22-SEP-26         1599 C003  Rahul Raj
      1004 23-SEP-26          699 C001  Arun Kumar
      1005 24-SEP-26          450 C004  Divya Kumar
      1006 24-SEP-26          750 C005  Karthik S
      1007 25-SEP-26         1200 C002  Aakash
      1008 25-SEP-26          550 C003  Rahul Raj
      1009 26-SEP-26          999 C005  Karthik S
      1010 24-SEP-26          550 C004  Divya Kumar
      1011 24-SEP-26          500 C004  Divya Kumar

  ORDER_ID ORDER_DAT TOTAL_AMOUNT CUSTO CUSTOMER_NAME
---------- --------- ------------ ----- -------------------------
      1012 24-SEP-26          550 C005  Karthik S
      1013 28-SEP-26          499 C002  Aakash
      1014 29-SEP-26          199 C003  Rahul Raj
      1015 30-SEP-26          229 C004  Divya Kumar
      1016 01-OCT-26          249 C005  Karthik S
      1017 27-SEP-26          449 C001  Arun Kumar

17 rows selected.
```

8. Find orders where customer details are missing

```
SQL> SELECT O.ORDER_ID, O.CUSTOMER_ID, O.ORDER_DATE
  2  FROM Customer C
  3  RIGHT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  4  WHERE C.CUSTOMER_ID IS NULL;

no rows selected
```

9. Display all payment records with order details

```
SQL> SELECT P.PAYMENT_ID, P.ORDER_ID, P.PAYMENT_MODE,
  2         P.PAYMENT_STATUS, O.ORDER_DATE, O.TOTAL_AMOUNT
  3  FROM Orders O
  4  RIGHT JOIN Payment P ON O.ORDER_ID = P.ORDER_ID
  5  ORDER BY P.PAYMENT_ID;

PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         1       1001 UPI                  Completed            20-SEP-26
        1299

         2       1002 Credit Card          Completed            21-SEP-26
         899

         3       1003 Debit Card           Completed            22-SEP-26
        1599


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         4       1004 UPI                  Pending              23-SEP-26
         699

         5       1005 Credit Card          Completed            24-SEP-26
         450

         6       1006 Cash                 Pending              24-SEP-26
         750
```

```
PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
         7       1007 UPI                  Completed            25-SEP-26
        1200

         8       1008 Debit Card           Completed            25-SEP-26
         550

         9       1009 UPI                  Pending              26-SEP-26
         999


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
        10       1010 Credit Card          Failed               24-SEP-26
         550

        11       1011 UPI                  Completed            24-SEP-26
         500

        12       1012 Credit Card          Completed            24-SEP-26
         550


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
        13       1013 UPI                  Completed            28-SEP-26
         499

        14       1014 Debit Card           Completed            29-SEP-26
         199

        15       1015 Credit Card          Completed            30-SEP-26
```

```
        14       1014 Debit Card           Completed            29-SEP-26
         199

        15       1015 Credit Card          Completed            30-SEP-26
         229


PAYMENT_ID   ORDER_ID PAYMENT_MODE         PAYMENT_STATUS       ORDER_DAT
---------- ---------- -------------------- -------------------- ---------
TOTAL_AMOUNT
------------
        16       1016 UPI                  Completed            01-OCT-26
         249

        17       1017 Cash                 Pending              27-SEP-26
         449


17 rows selected.
```

4. Retrieve Complete Order Details

10. Generate a complete order report

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME, OD.QUANTITY,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, Pay.PAYMENT_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  LEFT JOIN Payment Pay ON O.ORDER_ID = Pay.ORDER_ID
  8  ORDER BY O.ORDER_ID;

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Arun Kumar
Clear Shock Case
         2 20-SEP-26         1299 Completed

Arun Kumar
Soft Silicone Case
         1 20-SEP-26         1299 Completed

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------

Aakash
Crystal Clear Case
         1 21-SEP-26          899 Completed

Rahul Raj
Floral Printed Case
```

```
CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
         2 22-SEP-26         1599 Completed

Arun Kumar
Luxury Designer Case
         1 23-SEP-26          699 Pending

Divya Kumar

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Clear Shock Case
         1 24-SEP-26          450 Completed

Karthik S
Soft Silicone Case
         2 24-SEP-26          750 Pending


CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Aakash
Crystal Clear Case
         1 25-SEP-26         1200 Completed

Rahul Raj
Floral Printed Case
```

```
CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------

Karthik S
Luxury Designer Case
         1 26-SEP-26          999 Pending

Divya Kumar
Personal Photo Case

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
         1 24-SEP-26          550 Failed

Divya Kumar
Luxury Gold Case
         1 24-SEP-26          500 Completed

Karthik S

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Collage Photo Case
         1 24-SEP-26          550 Completed

Aakash
Heavy Duty Case
         1 28-SEP-26          499 Completed
```

```
CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
Rahul Raj
Matte Basic Case
         1 29-SEP-26          199 Completed

Divya Kumar
Classic Black Case
         1 30-SEP-26          229 Completed

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------

Karthik S
Slim Basic Case
         1 01-OCT-26          249 Completed

Arun Kumar
Personal Photo Case

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------------------
  QUANTITY ORDER_DAT TOTAL_AMOUNT PAYMENT_STATUS
---------- --------- ------------ --------------------
         1 27-SEP-26          449 Pending


18 rows selected.
```

5. Display Customer Purchase History

11. Display all products purchased by a customer

```
SQL> SELECT C.CUSTOMER_NAME, P.PRODUCT_NAME,
  2         O.ORDER_ID, OD.QUANTITY, OD.PRICE
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  JOIN Order_Details OD ON O.ORDER_ID = OD.ORDER_ID
  6  JOIN Product P ON OD.PRODUCT_ID = P.PRODUCT_ID
  7  WHERE C.CUSTOMER_ID = 'C001'
  8  ORDER BY O.ORDER_DATE DESC;

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------
Arun Kumar
Personal Photo Case
      1017          1        449

Arun Kumar
Luxury Designer Case
      1004          1        699

CUSTOMER_NAME
------------------------------
PRODUCT_NAME
--------------------------------------------------------------------
  ORDER_ID   QUANTITY      PRICE
---------- ---------- ----------

Arun Kumar
Clear Shock Case
      1001          2        499

Arun Kumar
Soft Silicone Case
```

12. Calculate total amount spent by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         NVL(SUM(O.TOTAL_AMOUNT), 0) AS TOTAL_SPENT
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_SPENT DESC;

CUSTO CUSTOMER_NAME                   TOTAL_SPENT
----- ------------------------------ ------------
C002  Aakash                               2598
C005  Karthik S                            2548
C001  Arun Kumar                           2447
C003  Rahul Raj                            2348
C004  Divya Kumar                          1729
C006  Priya Sharma                            0

6 rows selected.
```

13. Count the number of orders placed by each customer

```
SQL> SELECT C.CUSTOMER_ID, C.CUSTOMER_NAME,
  2         COUNT(O.ORDER_ID) AS NUMBER_OF_ORDERS
  3  FROM Customer C
  4  LEFT JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY NUMBER_OF_ORDERS DESC;

CUSTO CUSTOMER_NAME                 NUMBER_OF_ORDERS
----- ----------------------------- ---------------
C005  Karthik S                                   4
C004  Divya Kumar                                 4
C002  Aakash                                      3
C001  Arun Kumar                                  3
C003  Rahul Raj                                   3
C006  Priya Sharma                                0

6 rows selected.
```

14. Display the latest purchase details of each customer

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.TOTAL_AMOUNT, O.ORDER_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  WHERE O.ORDER_DATE = (
  6      SELECT MAX(O2.ORDER_DATE)
  7      FROM Orders O2
  8      WHERE O2.CUSTOMER_ID = C.CUSTOMER_ID
  9  )
 10  ORDER BY C.CUSTOMER_NAME;

CUSTOMER_NAME                 ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------------------------- ---------- --------- ------------
ORDER_STATUS
--------------------
Aakash                            1013 28-SEP-26          499
Delivered

Arun Kumar                        1017 27-SEP-26          449
Delivered

Divya Kumar                       1015 30-SEP-26          229
Delivered


CUSTOMER_NAME                 ORDER_ID ORDER_DAT TOTAL_AMOUNT
----------------------------- ---------- --------- ------------
ORDER_STATUS
--------------------
Karthik S                         1016 01-OCT-26          249
Delivered

Rahul Raj                         1014 29-SEP-26          199
Delivered
```

6. Multi-Table Business Reports

Report 1: Customer Order Report

```
SQL> SELECT C.CUSTOMER_NAME, O.ORDER_ID,
  2         O.ORDER_DATE, O.ORDER_STATUS
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  ORDER BY O.ORDER_ID;

CUSTOMER_NAME                  ORDER_ID ORDER_DAT ORDER_STATUS
------------------------------ ---------- --------- --------------------
Arun Kumar                           1001 20-SEP-26 Pending
Aakash                               1002 21-SEP-26 Shipped
Rahul Raj                            1003 22-SEP-26 Delivered
Arun Kumar                           1004 23-SEP-26 Pending
Divya Kumar                          1005 24-SEP-26 Shipped
Karthik S                            1006 24-SEP-26 Pending
Aakash                               1007 25-SEP-26 Delivered
Rahul Raj                            1008 25-SEP-26 Shipped
Karthik S                            1009 26-SEP-26 Pending
Divya Kumar                          1010 24-SEP-26 Shipped
Divya Kumar                          1011 24-SEP-26 Delivered

CUSTOMER_NAME                  ORDER_ID ORDER_DAT ORDER_STATUS
------------------------------ ---------- --------- --------------------
Karthik S                            1012 24-SEP-26 Pending
Aakash                               1013 28-SEP-26 Delivered
Rahul Raj                            1014 29-SEP-26 Delivered
Divya Kumar                          1015 30-SEP-26 Delivered
Karthik S                            1016 01-OCT-26 Delivered
Arun Kumar                           1017 27-SEP-26 Delivered

17 rows selected.
```

Report 2: Sales Report

```
SQL> SELECT P.PRODUCT_NAME,
  2         SUM(OD.QUANTITY) AS QUANTITY_SOLD,
  3         SUM(OD.QUANTITY * OD.PRICE) AS TOTAL_REVENUE
  4  FROM Product P
  5  JOIN Order_Details OD ON P.PRODUCT_ID = OD.PRODUCT_ID
  6  GROUP BY P.PRODUCT_ID, P.PRODUCT_NAME
  7  ORDER BY TOTAL_REVENUE DESC;

PRODUCT_NAME
--------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Floral Printed Case
            4          3599

Crystal Clear Case
            2          2099

Luxury Designer Case
            2          1698


PRODUCT_NAME
--------------------------------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Clear Shock Case
            3          1448

Soft Silicone Case
            3          1051

Personal Photo Case
            2           999
```

```
PRODUCT_NAME
----------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Collage Photo Case
            1           550

Luxury Gold Case
            1           500

Heavy Duty Case
            1           499


PRODUCT_NAME
----------------------------------------
QUANTITY_SOLD TOTAL_REVENUE
------------- -------------
Slim Basic Case
            1           249

Classic Black Case
            1           229

Matte Basic Case
            1           199


12 rows selected.
```

Report 3: Payment Analysis Report

```
SQL> SELECT PAYMENT_MODE,
  2         COUNT(*) AS NUMBER_OF_TRANSACTIONS,
  3         SUM(CASE WHEN PAYMENT_STATUS = 'Completed'
  4                  THEN 1 ELSE 0 END) AS SUCCESSFUL_PAYMENTS
  5  FROM Payment
  6  GROUP BY PAYMENT_MODE
  7  ORDER BY PAYMENT_MODE;

PAYMENT_MODE         NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
-------------------- ---------------------- ------------------
Cash                                      2                  0
Credit Card                               5                  4
Debit Card                                3                  3
UPI                                       7                  5
```

Report 4: Customer Purchase Analysis

- Top purchasing customers

```
SQL> SELECT C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  ORDER BY TOTAL_SPENDING DESC;

CUSTOMER_NAME                  TOTAL_SPENDING
------------------------------ --------------
Aakash                                   2598
Karthik S                                2548
Arun Kumar                               2447
Rahul Raj                                2348
Divya Kumar                              1729
```

- Customers with maximum orders

```
CUSTOMER_NAME                  NUMBER_OF_ORDERS
------------------------------ ----------------
Karthik S                                     4
Divya Kumar                                   4
Arun Kumar                                    3
Aakash                                        3
Rahul Raj                                     3
```

- Customers with highest spending

```
SQL> SELECT C.CUSTOMER_NAME,
  2         SUM(O.TOTAL_AMOUNT) AS TOTAL_SPENDING
  3  FROM Customer C
  4  JOIN Orders O ON C.CUSTOMER_ID = O.CUSTOMER_ID
  5  GROUP BY C.CUSTOMER_ID, C.CUSTOMER_NAME
  6  HAVING SUM(O.TOTAL_AMOUNT) = (
  7      SELECT MAX(TOTAL_SPENT)
  8      FROM (
  9          SELECT SUM(TOTAL_AMOUNT) AS TOTAL_SPENT
 10          FROM Orders
 11          GROUP BY CUSTOMER_ID
 12      )
 13  );

CUSTOMER_NAME                 TOTAL_SPENDING
----------------------------- --------------
Aakash                                  2598
```